

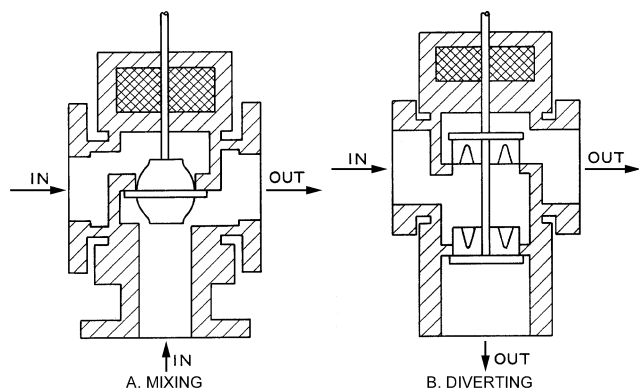


Fig. 8 Typical Three-Way Mixing and Diverting Globe Valves

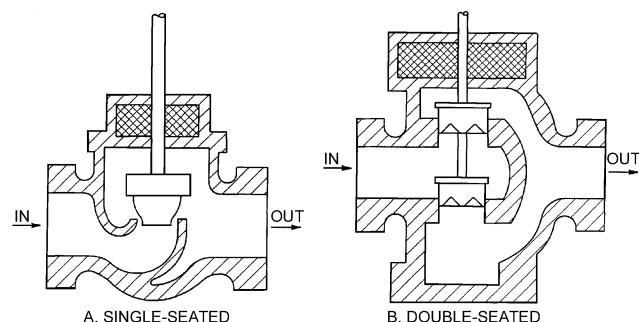


Fig. 9 Typical Single- and Double-Seated Two-Way Globe Valves

to both outlets. Three-way diverting valves are more expensive and have more complex applications, and generally are not used in typical HVAC systems.

A **two-way globe valve** may be either single or double seated. A **single-seated valve** (Figure 9A) is designed for tight shutoff. Appropriate disk materials for various pressures and media are used. A **double-seated** or **balanced valve** (Figure 9B) is designed so that the media pressure acting against the valve disk is essentially balanced, reducing the actuator force required. It is widely used where fluid pressure is too high to allow a single-seated valve to close or to modulate properly. It is not usually used where tight shutoff is required.

A **butterfly valve** consists of a heavy ring enclosing a disk that rotates on an axis at or near its center and is similar to a round single-blade damper. In principle, the disk seats against a ring machined within the body or a resilient liner in the body. Two butterfly valves can be used together to act like a three-way valve for mixing or diverting. In this arrangement, one valve is set up as normally open and the other is normally closed. In applications with pipe sizes 100 mm and above, butterfly valves are either two position or modulating, and they are less expensive than globe-style valves (see Chapter 47 of the 2016 *ASHRAE Handbook—HVAC Systems and Equipment* for information on globe valves).

A **ball valve** consists of a ball with a hole drilled through it, rotating in a valve body. Ball valves are increasingly popular because of their low cost and high close-off ratings. Features that provide flow characteristics similar to or better than globe valves are available.

Pressure-independent valves are control valves with integral pressure regulators. This allows the valve to maintain a constant flow at a given shaft position because the integral pressure regulator maintains a constant differential pressure across the valve's orifice, regardless of system pressure fluctuations.

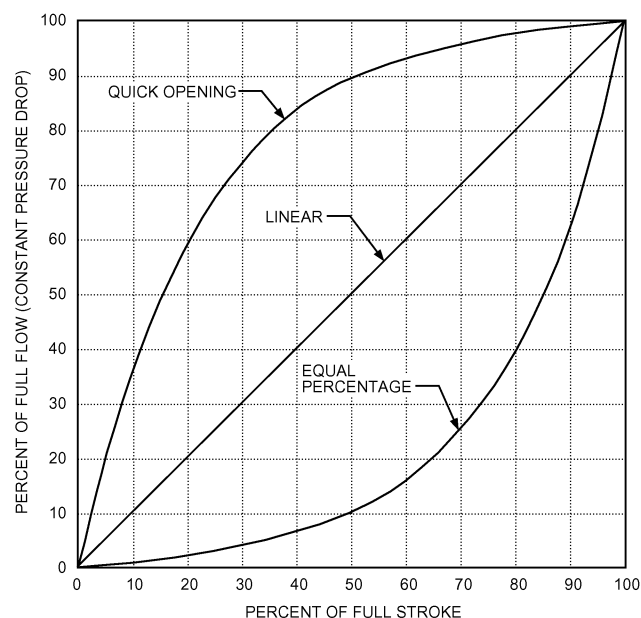


Fig. 10 Typical Flow Characteristics of Valves

Flow Characteristics. Valve performance is expressed in terms of its flow characteristics as it operates through its stroke, based on a constant pressure drop. Three common characteristics are shown in Figure 10 and are defined as follows:

- **Quick opening.** Maximum flow is approached rapidly as the device begins to open.
- **Linear.** Opening and flow are related in direct proportion.
- **Equal percentage.** Each equal increment of opening increases flow by an equal percentage over the previous value.

On a pressure-dependent valve, because pressure drop across the valve's orifice seldom remains constant as its opening changes, actual performance usually deviates from the published characteristic curve. The magnitude of deviation is determined by the overall design. For example, in a system arranged so that control valves or dampers can shut off all flow, pressure drop across a controlled device increases from a minimum at design conditions to total pressure drop at no flow. Figure 11 shows the extent of resulting deviations for a valve or damper designed with a linear characteristic, when selection is based on various percentages of total system pressure drop. To allow for adequate control by the valve, design pressure drop should be a reasonably large percentage of total system pressure drop at the valve (i.e., adequate valve authority) and the system should be designed and controlled so that pressure drop remains relatively constant. Hydronic piping circuits and valve authority are discussed in Chapters 13 and 47 of the 2016 *ASHRAE Handbook—HVAC Systems and Equipment*, respectively.

Selection and Sizing. Higher pressure drops for control devices are obtained by using smaller sizes, with a possible increase in size of other equipment in the system. Sizing control valves is discussed in Chapter 47 of the 2016 *ASHRAE Handbook—HVAC Systems and Equipment*.

Steam Valves. Steam-to-water and steam-to-air heat exchangers are typically controlled by regulating steam flow using a two-way throttling valve. One-pipe steam systems require a line-size, two-position valve for proper condensate drainage and steam flow; two-pipe steam systems can be controlled by two-position or modulating (throttling) valves. Maximum pressure drop for steam valves is a function of operating pressure and cannot be exceeded.

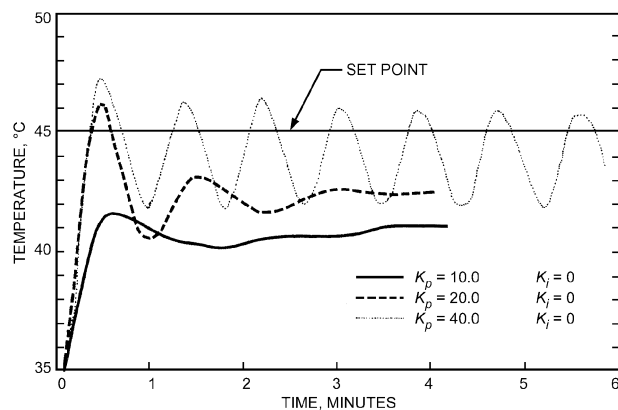


Fig. 21 Response of Discharge Air Temperature to Step Change in Set Points at Various Proportional Constants with No Integral Action

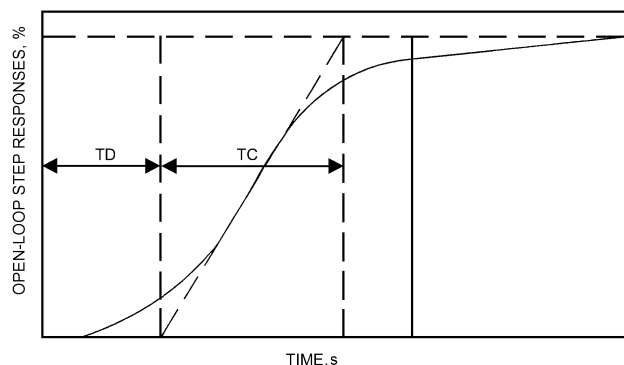


Fig. 22 Open-Loop Step Response Versus Time

- Record the proportional band as PB_u and the period of oscillations as T_u .
- Use the recorded proportional band and oscillation period to calculate controller settings as follows:

Proportional only:

$$PB = 1.8(PB_u) \text{ percent} \quad (4)$$

Proportional plus integral (PI):

$$PB = 2.22(PB_u) \text{ percent} \quad (5)$$

$$T_i = 0.83T_u \text{ minute per repeat} \quad (6)$$

Proportional plus integral plus derivative (PID):

$$PB = 1.67(PB_u) \text{ percent} \quad (7)$$

$$T_i = 0.50T_u \text{ minute per repeat} \quad (8)$$

$$T_d = 0.125T_u \text{ minute} \quad (9)$$

First-Order-plus-Dead-Time (Open-Loop) Method. The open-loop method introduces a step change in input into the opened control loop. A graphical technique is used to estimate the process transfer function parameters. Proportional and integral terms are calculated from the estimated process parameters using a series of equations.

The value of the process variable must be recorded over time, and the dead time and time constant must be determined from it. This can be accomplished graphically, as seen in Figure 22. The first-order-plus-dead-time method is as follows:

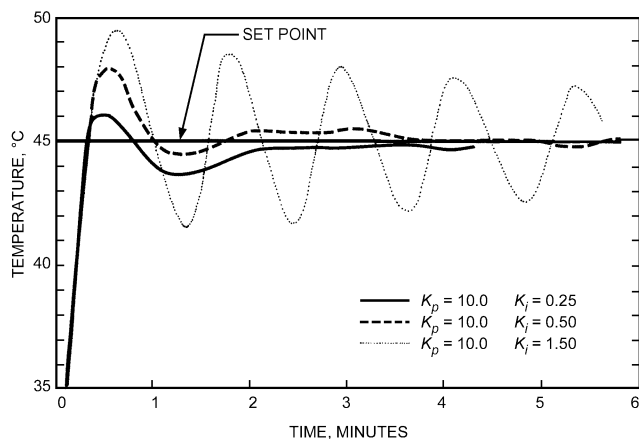


Fig. 23 Response of Discharge Air Temperature to Step Change in Set Points at Various Integral Constants with Fixed Proportional Constant

- Adjust controller manual output to give a midscale measurement.
- Arrange to record the process variable over time.
- Move the manual output of the controller by 10% as rapidly as possible to approximate a step change.
- Record the value of the process variable over time until it reaches a new steady-state value.
- Determine dead time and time constant.
- Use dead time (TD) and time constant (TC) values to calculate PID values as follows:

$$\text{Gain} = \frac{\% \text{ change in controlled variable}}{\% \text{ change in control signal}} \quad (10)$$

Proportional only:

$$PB = \text{Gain}/(TC/TD) \quad (11)$$

Proportional plus integral (PI):

$$PB = 0.9(\text{Gain})/(TC/TD) \quad (12)$$

$$T_i = 3.33(TD) \quad (13)$$

Proportional-integral-derivative (PID):

$$PB = 1.2(\text{Gain})/(TC/TD) \quad (14)$$

$$T_i = 2(TD) \quad (15)$$

$$T_d = 0.5(TD) \quad (16)$$

Trial and Error. This method involves adjusting the gain of the proportion-only controller until the desired response to a set point is observed. Conservative tuning dictates that this response should have a small initial overshoot and quickly damp to steady-state conditions. Set-point changes should be made in the range where controller saturation, or output limit, is avoided. The integral term is then increased until changes in set point produce the same dynamic response as the controller under proportional control, but with the response now centered about the set point (Figure 23).

Tuning Digital Controllers

In tuning digital controllers, additional parameters may need to be specified. The digital controller sampling interval is critical because it can introduce harmonic distortion if not selected properly. This sampling interval is usually set at the factory and may not be adjustable. A controller sampling interval of about one-tenth of the controlled-process time constant usually provides adequate control. Many digital control algorithms include an error dead band to eliminate unnecessary control actions when the process is near set

point. Hysteresis compensation is possible with digital controllers, but it must be carefully applied because overcompensation can cause continuous cycling of the control loop.

Computer Modeling of Control Systems

Each component of a control system can be represented by a transfer function, which is an idealized mathematical representation of the relationship between the input and output variables of the component. The transfer function must be sufficiently detailed to cover both the dynamic and static characteristics of the device. The dynamics are represented in the time domain by a differential equation. In environmental control, the transfer function of many of the components can be adequately described by a first-order differential equation, implying that the dynamic behavior is dominated by a single capacitance factor. For a solution, the differential equation is converted to its Laplace or z-transform.

For more information on computer modeling programs, see Chapter 40 of the 2015 *ASHRAE Handbook—HVAC Applications*.

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Applications of this include ceiling fans and personal environmental control systems in offices and transportation systems.

Vertical Air Temperature Difference

In most buildings, air temperature normally increases with height above the floor. If the gradient is sufficiently large, local warm discomfort can occur at the head and/or cold discomfort can occur at the feet, although the body as a whole is thermally neutral. Among the few studies of vertical air temperature differences and the influence of thermal comfort reported are Eriksson (1975), McNair (1973), McNair and Fishman (1974), and Olesen et al. (1979). Subjects were seated in a climatic chamber and individually exposed to different air temperature differences between head and ankles (Olesen et al. 1979). During the tests, the subjects were in thermal neutrality because they were allowed to change the temperature level in the test room whenever they desired; the vertical temperature difference, however, was kept unchanged. Subjects gave subjective reactions to their thermal sensation; Figure 12 shows the percentage of dissatisfied as a function of the vertical air temperature difference between head (1.1 m above the floor) and ankles (0.1 m above the floor).

A head-level air temperature lower than that at ankle level is not as critical for occupants. Eriksson (1975) indicated that subjects could tolerate much greater differences if the head were cooler. This observation is verified in experiments with asymmetric thermal radiation from a cooled ceiling (Fanger et al. 1985).

Warm or Cold Floors

Because of direct contact between the feet and the floor, local discomfort of the feet can often be caused by a too-high or too-low floor temperature. Also, floor temperature significantly influences a room's mean radiant temperature. Floor temperature is greatly affected by building construction (e.g., insulation of the floor, above a basement, directly on the ground, above another room, use of floor heating, floors in radiant-heated areas). If a floor is too cold and the

occupants feel cold discomfort in their feet, a common reaction is to increase the temperature level in the room; in the heating season, this also increases energy consumption. A radiant system, which radiates heat from the floor, can also prevent discomfort from cold floors.

The most extensive studies of the influence of floor temperature on foot comfort were performed by Olesen (1977a, 1977b), who, based on his own experiments and reanalysis of data from Nevins and Feyerherm (1967), Nevins and Flinner (1958), and Nevins et al. (1964), found that flooring material is important for people with bare feet (e.g., in swimming halls, gymnasiums, dressing rooms, bathrooms, bedrooms). Ranges for some typical floor materials are as follows:

Textiles (rugs)	21 to 28°C
Pine floor	22.5 to 28°C
Oak floor	24.5 to 28°C
Hard linoleum	24 to 28°C
Concrete	26 to 28.5°C

To save energy, insulating flooring materials (cork, wood, carpets), radiant heated floors, or floor heating systems can be used to eliminate the desire for higher ambient temperatures caused by cold feet. These recommendations should also be followed in schools, where children often play directly on the floor.

For people wearing normal indoor footwear, flooring material is insignificant. Olesen (1977b) found an optimal temperature of 25°C for sedentary and 23°C for standing or walking persons. At the optimal temperature, 6% of occupants felt warm or cold discomfort in the feet. Figure 13 shows the relationship between floor temperature and percent dissatisfied, combining data from experiments with seated and standing subjects. In all experiments, subjects were in thermal neutrality; thus, the percentage of dissatisfied is only related to discomfort caused by cold or warm feet. No significant difference in preferred floor temperature was found between females and males.

8. SECONDARY FACTORS AFFECTING COMFORT

Temperature, air speed, humidity, their variation, and personal parameters of metabolism and clothing insulation are primary factors that directly affect energy flow and thermal comfort. However, many secondary factors, some of which are discussed in this section, may more subtly influence comfort.

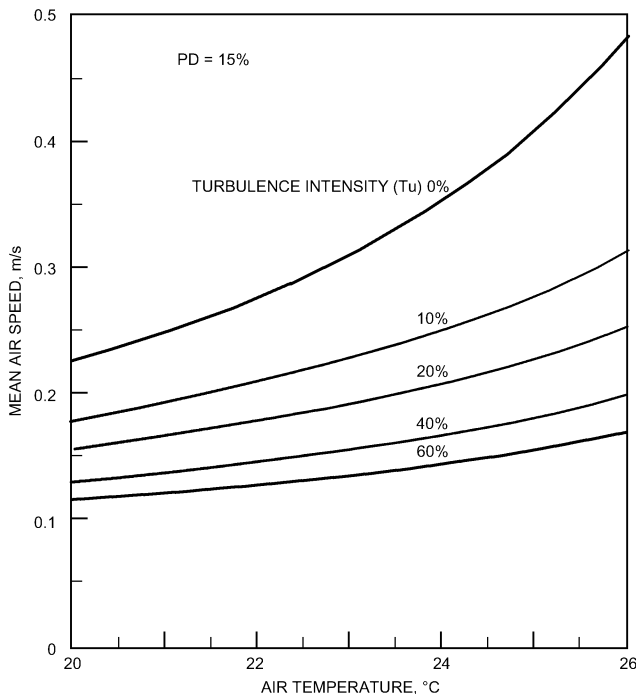


Fig. 11 Draft Conditions Dissatisfying 15% of Population (PD = 15%)

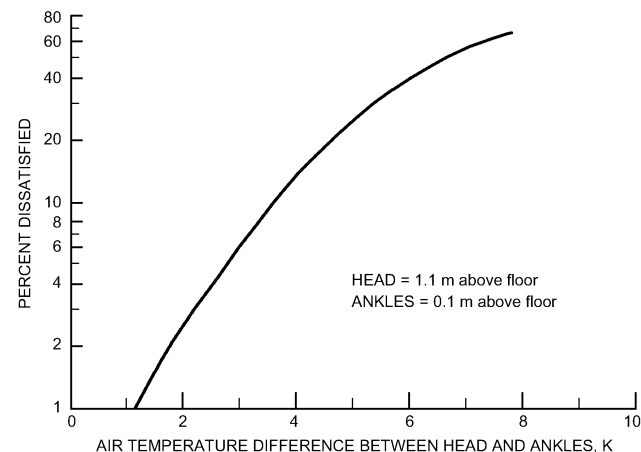


Fig. 12 Percentage of Seated People Dissatisfied as Function of Air Temperature Difference Between Head and Ankles

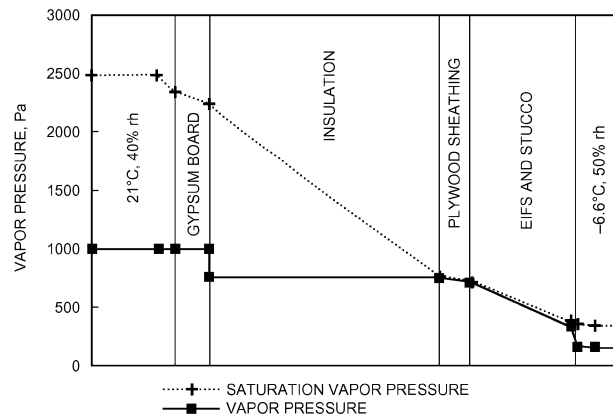


Fig. 11 Drying Wet Sheathing, Winter (Example 9)

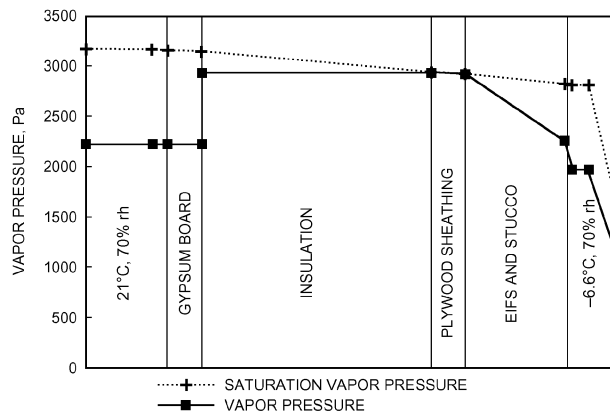


Fig. 12 Drying Wet Sheathing, Summer (Example 9)

For many applications and for design guide development, actual behavior of an assembly under transient climatic conditions may be simulated to account for short-term processes such as driving rain absorption, summer condensation, and phase changes. Computer simulations allow designers to model these conditions over time. It is important, however, to understand the model's application limits.

Applying one of these models requires at least the following information: exterior climate conditions, indoor temperature and humidity, and building assembly materials and sizes. Many programs include a material property database and exterior climate and indoor condition data, allowing simple modeling to be performed without customization. However, using generic material property and weather data may not accurately recreate actual target conditions.

Features of a complete moisture analysis model include

- Transient heat, air, and moisture transport formulation, incorporating the physics of
 - Airflow
 - Water vapor transport by advection (combination of water vapor diffusion and air-driven vapor flow)
 - Liquid transport by capillary action, gravity, and pressure differences
 - Heat flow by apparent conduction, convection, and radiation
 - Heat and moisture storage/capacity of materials
 - Condensation and evaporation processes with linked latent-to-sensible heat transformation
 - Freezing and thawing processes with linked latent-to-sensible heat transformation and based on laws of conservation of heat, mass, and momentum

- Material properties as functions of moisture content, relative humidity, and temperature, such as
 - Density
 - Air properties: permeability and permance
 - Thermal properties: specific heat capacity, apparent thermal conductivity, Nusselt numbers, and long-wave emittance (for cavities and air spaces)
 - Moisture properties: porosity, sorption curve, water retention curve, vapor permeability, water permeability, or liquid diffusivity
- Boundary conditions (generally on an hourly basis)
 - Outside temperature and relative humidity
 - Incident short-wave solar and long-wave sky radiation (depending on inclination and orientation)
 - Wind speed, orientation, and pressures
 - Wind-driven rain at exterior surfaces (depending on location and aerodynamics)
 - Interior temperature, air pressure excess, relative humidity (or interior moisture sources and ventilation flows), and air stratification
 - Surface conditions
 - Heat transfer film coefficients (combined convection and radiation, separate for convection and radiation)
 - Mass transfer film coefficients
 - Short-wave absorptance of exterior surfaces
 - Long-wave emittance of exterior surfaces
 - Contact conditions between layers and materials. Interfaces may be bridgeable for vapor diffusion, airflow, and gravity or pressure liquid flow only. They may be ideally capillary, introduce additional capillary resistance, or behave as real contact.

Not all these features are required for every analysis, and some applications may need additional information (e.g., moisture flow through unintentional cracks and intentional openings, rain penetration through veneer walls and exterior cladding). To model these phenomena accurately, experiments may be needed to define systems and subsystems in field situation, because only then are all exterior loads and influences captured.

It is important to recognize that simulation results are based on input data. Therefore, the more accurate the input data used, the closer the results will match real-world conditions. Exterior weather conditions, material properties, and interior operating conditions all vary widely, so it is important to get the best data available on materials being modeled. For most users, this is a difficult task. Many product manufacturers do not provide the material property data needed for the simulations. Developing weather data for a particular site is also beyond the expertise of many.

Combined heat, air, and moisture models also have limitations. Users should be aware which transport phenomena and types of boundary conditions are included and which are not. For instance, some models cannot handle air transport or rain wetting of the exterior. Even an apparently simple problem, such as predicting rain leakage through a brick veneer, is beyond existing tools' capabilities. In such cases, simple qualitative schemes and field tests still are the way to proceed. In addition, results also tend to be very sensitive to the choice of indoor and outdoor conditions. Usually, exact conditions are not known.

Outputs from these programs typically include the moisture content of materials as well as relative humidity within the assembly. Interpretation of results is not easy: accurate data on moisture and temperature conditions that materials can tolerate are often not available. Although moisture accumulation may result in indoor air quality issues or material degradation, the effect of moisture accumulation in building assemblies depends on many factors, including choice of construction materials, and varies by building, making interpretation of results even more difficult.

4. AIR MOVEMENT

Moisture movement through building envelope assemblies is more strongly affected by air movement than by diffusion. To minimize moisture penetration by air leakages, the building envelope should be as airtight as possible. The airflow retarder must also be sufficiently strong and well supported to resist wind loads.

In older residential buildings, air leakage provided sufficient ventilation and rarely led to interstitial condensation. However, in airtight buildings, mechanical ventilation is needed to ensure acceptable air quality and prevent moisture and health problems caused by excessive indoor humidity. Ventilation of the wall assembly and/or drainage must go to the outside of the airtight layer of construction, or it will increase building air leakage. To avoid moisture problems at the airtight layer, either the layer temperature must be kept above the dew point by locating it on the warm side of the insulation, or the layer permeance must allow vapor transmission.

As described in detail in the section on Leakage Distribution in Chapter 16, air leakage through building envelopes is not confined to doors and windows. Although 6 to 22% of air leakage occurs there, 18 to 50% typically takes place through walls, and 3 to 30% through the ceiling. Leakage often occurs between sill plate and foundation, through interior walls, electrical outlets, plumbing penetrations, and cracks at top and bottom of exterior walls.

Not all cracks and openings can be sealed in existing buildings, nor can absolutely tight construction be achieved in new buildings. Provide as tight an enclosure as possible to reduce leakage, minimize potential condensation within the envelope, and reduce energy loss. However, the project team must also recognize the impact of airtightness.

Moisture accumulation in building envelopes can also be minimized by controlling the dominant direction of airflow by operating the building at a small negative or positive air pressure, depending on climate. In cooling climates, pressure should be positive to keep out humid outside air. In heating climates, pressure should be neither strongly negative, which could risk drawing soil gas or combustion products indoors and affect occupant comfort, nor strongly positive, which could risk driving moisture into building envelope cavities.

Equivalent Permeance

Dew-point analysis allows simple estimation of the effect of wall and roof cavity ventilation on heat and vapor transport by using parallel thermal and vapor diffusion resistances (TenWolde and Carll 1992; Trethowen 1979). These parallel resistances account for heat and vapor that bypass exterior material layers with ventilation air from outside. Equivalent thermal and water vapor diffusion resistances are approximated from the following equations:

$$R_{par} = \frac{1}{m_a c}$$

$$Z_{par} = \frac{1}{m_a \xi}$$

where

R_{par} = parallel equivalent thermal resistance, (m²·K)/W

Z_{par} = parallel equivalent water vapor diffusion vapor flow resistance, (Pa·s·m²)/ng

m_a = ventilation flux, kg/(m²·s)

ξ = 0.62/ P_a , where P_a is atmospheric pressure, Pa

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